

Linear Algebra

Selected Topics on Mathematical Physics | Autumn 2026

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I Prerequisites

Students are expected to be familiar with the following topics:

- Elementary complex analysis
- Elementary calculus

II Course Structure

The course meets twice per week for lectures. In addition, a weekly tutoring session is held to support work on the problem sheets; attendance is strongly encouraged, as the sheets form an integral part of the course.

	Format	Duration
Regular sessions	Theory and derivations	80 minutes
Tutoring session	Problem sheet workshop	90 minutes

Problem sheets should be attempted *before* the tutoring session. Hints and solutions are discussed in the tutoring session and collected shortly thereafter.

III Course Description

Linear algebra is the mathematical study of vector spaces, linear transformations, and the structures that emerge from them — a language so important in modern mathematics and its applications that it is impossible to go far in any quantitative discipline without it. This course develops the theory from its foundations, using tools from modern algebra such as group theory to formally construct the defining properties of a linear space. The use of group theory further extends to define the symmetric group S_n and thus define multilinear alternating maps between an n -fold of a linear space and another to derive one of the most fundamental tools in linear algebra, namely, determinant functions.

The reach of linear algebra extends across virtually every corner of science and mathematics. It is the native language of quantum mechanics, the engine behind numerical methods and data science, the foundation of differential equations and functional analysis, and the framework within which Fourier analysis and signal processing are properly understood. Students who master this course will find that it reorganises and clarifies much of what they already know — and opens the door to nearly everything that comes next.

IV Contents

1. **Linear Spaces** — Definition of linear space via abstract algebra; bases; inner products; linear space homomorphisms; dual spaces; multilinear maps; determinant functions.
2. **Algebras** — Algebras from linear spaces; algebra homomorphisms; division algebras; the algebra of quaternions; idempotents; polynomial algebra.
3. **Operator Algebra** — Algebra of $\text{End}(\mathcal{V})$; functions of operators; derivatives of operators; hermitian operators; unitary operators; projection operators.
4. **Matrices** — Representation of vectors and operators; matrix arithmetic; orthonormal bases; change of basis; determinant of a matrix; inverse of a matrix.
5. **Spectral Decomposition** — Eigenvalues and eigenvectors; complex spectral decomposition; diagonalisation; polar decomposition.

V References

The star rating reflects the mathematical sophistication of each text: (★) accessible with the course prerequisites; (★★) requires additional mathematical maturity; (★★★) intended for readers revisiting the subject — not recommended if this is your first encounter with the material.

- [1] Hassani, S. *Mathematical Physics: A Modern Introduction to Its Foundations*. 2nd ed. Springer, 2013. [**Primary (★★)**]
- [2] Szekeres, P., *A Course in Modern Mathematical Physics: Groups, Hilbert Space and Differential Geometry*. Cambridge University Press, 2004. [**Background reference (★★)**]